

Effects of Feedback-Based Reinforcement in Human Sequential Decision-Making

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1 Introduction

Reinforcement learning (RL) is a subfield of machine learning and AI which involves training agents via trial and error, where agents are rewarded or penalized based their actions. In [1], the paper shows that RL agents procrastinate but still try to complete a task when incentivized to complete it with a positive terminal reward, but disincentivized to contribute to the subtasks that lead up to it. This demonstrates the idea that negative rewards lead to discouraging behaviour, while positive rewards provide a motivating factor.

In this project, we are motivated to investigate the effect of positive and negative feedback on sequential decision-making task performance from a human factors perspective, where we see if the studied behaviour of AI agents in [1] may also apply to humans through statistical means. The purpose of this study is to examine whether feedback valence (positive vs. negative) affects response time and accuracy in a rule-discovery task for 2T8 EngSci students. It is hypothesized that participants receiving positive feedback will demonstrate faster response times but lower accuracy compared to those receiving negative feedback.

2 Methods

Participants: The population studied consisted of only 2T8 EngSci students. 30 students were selected as participants through simple random sampling, implemented through a numbered names list of 2T8 EngSci students and a random number generator.

Apparatus: The apparatus consisted of a web app (see Appendix 7.2; for quick reference, see Appendix 7.3), with 10 slots in a cellphone dial format to display user input. Slots are filled left to right by clicking buttons. A 10-minute timer was displayed on the screen with seconds as the smallest time step.

Task: The task for the participant was to figure out the hidden rule that they must click the numbers on the dial in an alternating highest-lowest number sequence, without repeating digits (i.e. 9-0-8-1-7-2-6-3-5-4). They must correctly enter that sequence into the 10 slots.

Procedure: Participants were split through a numbered names list and a random number generator. The first 15 generated participants were assigned Type A, and the others Type B. Participants were invited to a room with minimal noise and visual distractions. They were seated in front of a laptop with the web app open for their group type. They were free to click the begin button when they felt ready, and the apparatus screen would appear for them to interact with. The user input sequence was automatically checked and reset when the 10 slots were filled. The task continued until either the timer ran out or the participant had entered the correct sequence as outlined by the Task section. When the task finished,

the participant was asked to complete the survey to measure the covariates.

Independent variable: Type A (Positive Feedback): Every time the participant entered a correct number into a slot, a pleasant sound was played, and the slot lit light up green. Type B (Negative Feedback): Every time the participant entered an incorrect number into a slot, an unpleasant sound was played, and the slot lit up red.

Dependent variables: The dependent variables were thinking time and accuracy. Thinking time was recorded as the total time a participant took to complete the task, subtracted by the time that feedback was given (0.25 seconds for feedback), with a maximum of 10 minutes.

$$t_{thinking} = t_{total} - (0.25 \times x), \quad x = \begin{cases} \text{correct clicks,} & \text{if feedbackType = POSITIVE} \\ \text{incorrect clicks,} & \text{if feedbackType = NEGATIVE} \end{cases} \quad (1)$$

Accuracy was measured by calculating the number of correct clicks divided by the total clicks:

$$\text{accuracy} = \frac{\text{correct clicks}}{\text{total clicks}} \quad (2)$$

Covariate variables: Two covariates were measured, which were the participants' sleep duration the night before, as well as their memorization ability. First, the participant was asked to estimate their sleep duration the night before. Second, they were to complete a given number memory game from the Human Benchmark website. Participants were shown sequences of increasing lengths and asked to repeat them. The longest correctly recalled sequence was recorded as the memory score. The same sequence was used for all participants.

Biases and Limitations: Findings may not perfectly represent the effect of positive and negative feedback on other sequential decision-making tasks because only one puzzle type is used. This experiment was conducted with only EngSci 2T8 students, so findings may not be shared in wider populations. The limited sample size of 30 is also a concern, as problem-solving ability may dominate the statistical findings, making different feedback types less impactful. Although the random sampling method reduced bias and helped balance demographics, some imbalance may still have existed, given the relatively small sample size.

3 Analysis

The main purpose of the analysis is to determine whether feedback type (positive vs. negative) has significant effects on the dependent variables, accuracy and response time, while accounting for potential covariates.

3.1 Descriptive Statistics

Descriptive statistics are first generated to observe differences in the sample groups, which includes **mean**, **median**, **standard deviation**, and **range**, presented in **box plots** and **bar charts**. The box plot will provide a summary of performance (represented by the median and whisker positions) versus variability (spread of data represented by the interquartile ranges) for both thinking time and accuracy. Since the box plot does not clearly demonstrate the means, a bar chart with error bars will be employed to demonstrate average performances and variability between the two feedback types.

3.2 Inferential Statistics

In order to conduct a T-test for response time and accuracy, the normality of the response time and accuracy samples must be verified using **Quantile-Quantile (QQ) plots**. Then, it is assumed that the feedback type (positive or negative) has no effect on response time or accuracy. Hence, the null hypotheses are defined as:

$$H_{0,time} : \mu_{P,time} - \mu_{N,time} = 0, \quad (3)$$

$$H_{0,acc} : \mu_{P,acc} - \mu_{N,acc} = 0 \quad (4)$$

To determine if positive or negative feedback has a statistically significant effect on response time or accuracy, the alternative hypothesis must be proven:

$$H_{1,time} : \mu_{P,time} - \mu_{N,time} \neq 0 \quad (5)$$

$$H_{1,acc} : \mu_{P,acc} - \mu_{N,acc} \neq 0 \quad (6)$$

Since a difference is being tested for and the population variances of positive and negative groups are not known, the two-tailed two-sample T-test is used. Specifically, the **Welch T-test** is used as the samples distributions of the positive and negative sample groups are not assumed to be the same. The significance level is set reasonably at $\alpha = 0.05$. If the p-values from the t-test are less than α , the null hypotheses is refuted which suggests a statistically significant difference between the two feedback groups. Then, the mean response time and or accuracy between feedback groups can then be compared to determine the direction of difference.

3.3 Regression Analysis

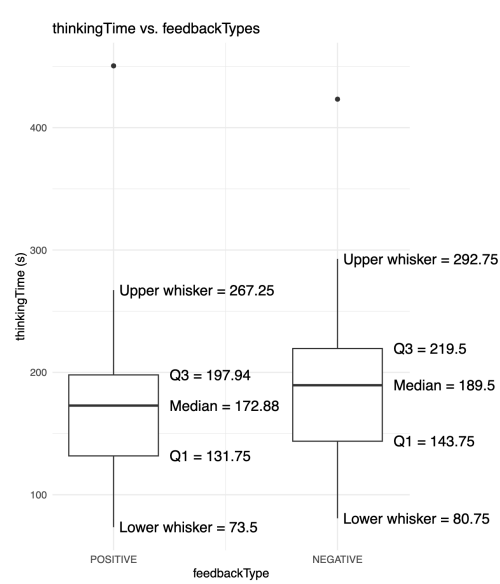
In order to account for the covariates, **multiple linear regression** is performed with positive feedback as the reference, and negative feedback, memory score and sleep time as the predictors. The null hypothesis is that for each predictor the coefficient is 0, indicating that there is no effect on the dependent variables. Normality of the residuals of the dependent variables is checked with QQ plots again.

3.4 Additional Analysis

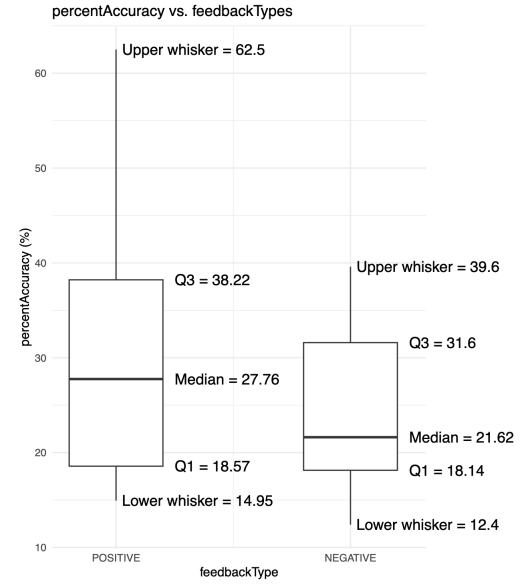
To analyze the time-accuracy tradeoff, a **linear fit** will be applied to the data to determine a possible relationship between these two metrics. Additionally, a **residuals plot** will be to analyze the quality and error of the linear fit.

4 Results

4.1 Mean and Median Analysis of Hypothesis



(a) The box plot of thinking time versus feedback types.



(b) The box plot of accuracy versus feedback types.

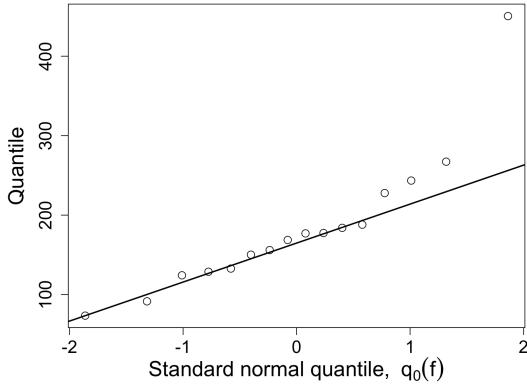
Figure 1: Box plots of thinking time and accuracy across feedback types.

In Figures 7a and 7b, it is shown that all median and quartile ranges (i.e. 50% of the data) were faster than the negative box for the thinking time, and are greater for the accuracy. This suggests that the initial hypothesis is partly correct; participants receiving positive feedback had faster response times, but they also had a higher accuracy, which was not initially

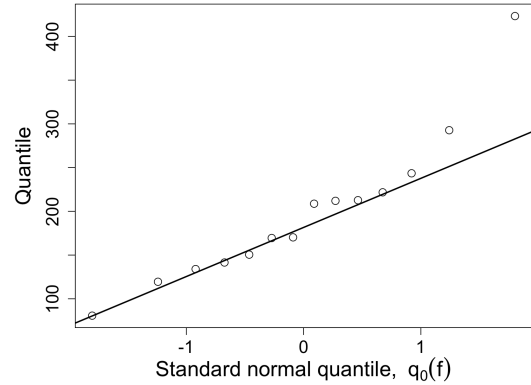
predicted. This is also demonstrated in the bar chart displaying the means of variables (shown in Appendix 7.5), where the mean thinking time is faster and mean accuracy is greater for the positive feedback type.

Initially, the accuracy metric was hypothesized to be worse for positive feedback because less information is provided than negative feedback (i.e. it is more likely that people click incorrect buttons than correct ones). However, positive feedback may have provided better accuracy due to the idea that positive reinforcement leads to motivating behaviour (as described in [1]). In the same way, this could explain the faster response times observed, as positive feedback influences greater motivation to complete the task more quickly.

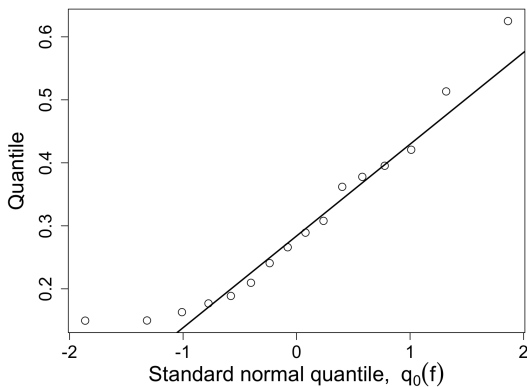
4.2 Determining Normality of Response Time and Accuracy



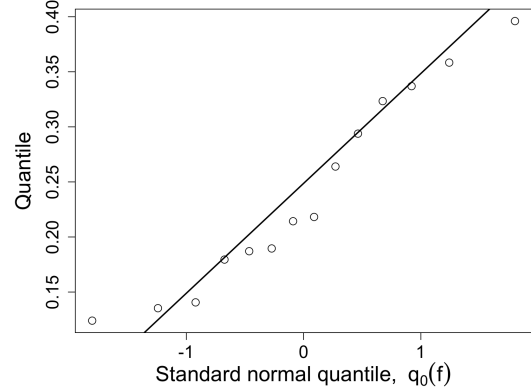
(a) QQ plot of response time for the positive feedback group.



(b) QQ plot of response time for the negative feedback group.



(c) QQ plot of accuracy for the positive feedback group.



(d) QQ plot of accuracy for the negative feedback group.

Figure 2: QQ plots for response time and accuracy of both positive and negative feedback groups.

The trend for the response time plots (2b,2a) appears to be more linear than the accuracy plots (2d,2c). This is expected as accuracy is inherently bounded between 0 and 1, which is not characteristic of normally distributed data. Regardless, all four accuracy and response time plots appear to exhibit nearly straight-line trends, indicating that the data can be approximated as normal. The T-test is therefore applicable to response time and accuracy data for both positive and negative feedback groups.

4.3 Welch T-test on Response Time and Accuracy

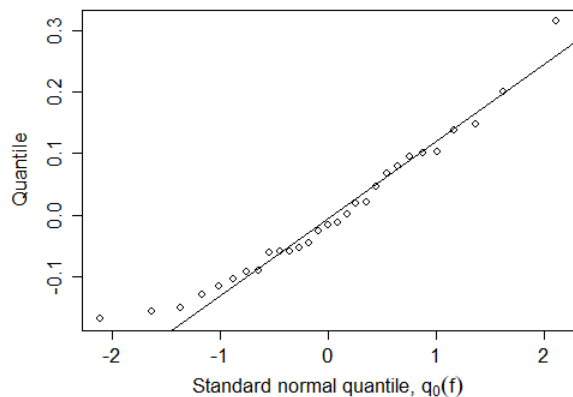
Metric	Response Time	Accuracy
Mean (Positive)	183.828	0.3022
Mean (Negative)	198.625	0.240
<i>t</i> -statistic	0.468	-1.477
Degrees of Freedom (<i>df</i>)	27.690	25.801
<i>p</i> -value	0.643	0.152
95% CI	[-49.948, 79.541]	[-0.149, 0.024]

Table 1: Results of Welch two-sample t-tests comparing response time and accuracy between positive and negative feedback groups.

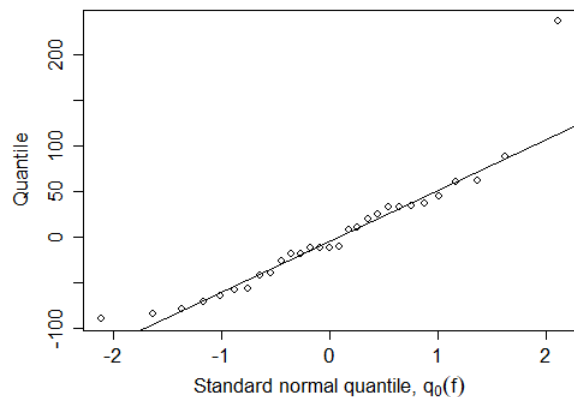
Referencing Table 1, the mean response time of the positive feedback group (183.828) is lower than the negative feedback group (198.625). The mean accuracy of the positive feedback group is higher (0.3022) than the negative feedback group (0.240).

However, since the *p*-values for response time (0.643) and accuracy (0.152) are greater than the selected α of 0.05, the null hypotheses $H_{0,time}$ and $H_{0,acc}$ are not rejected. In other words, there is insufficient statistical evidence to determine if feedback type affects response time or accuracy in the population of 2T8 Engineering Science students.

4.4 Determining Normality of Residuals of Response Time and Accuracy



(a) QQ plot of residuals for the accuracy variable.



(b) QQ plot of residuals for the response time variable.

Figure 3: QQ plots for residuals of response time and accuracy for multiple linear regression.

As shown in the accuracy and response time residual QQ plots (3a,3b), the residuals for both models show an overall linear trend, indicating approximate normality of the residuals. Thus, multiple linear regression is applicable for both dependent variables as the normality assumption is reasonably satisfied.

4.5 Covariates Control and Multiple Linear Regression Analysis

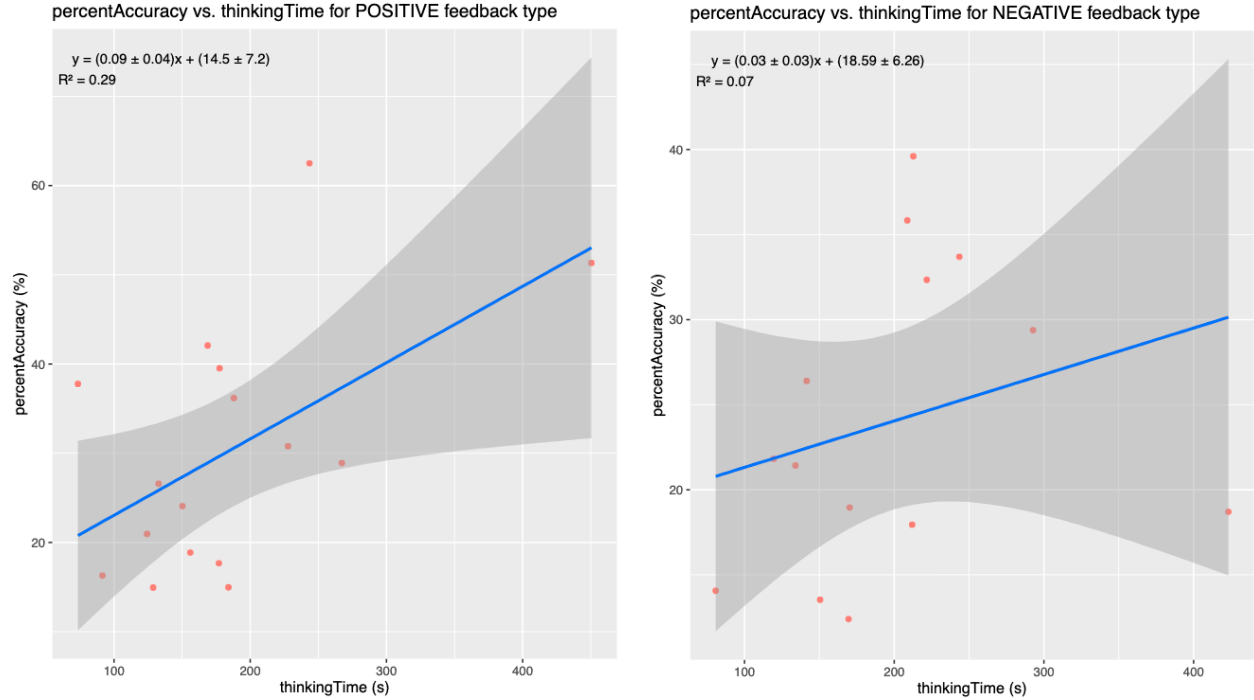
Predictor	Estimate	p-value
<i>Accuracy Model</i>		
Feedback (Negative)	-0.057	0.224
Sleep Time	-0.010	0.456
Memory Score	-0.003	0.702
<i>Response Time Model</i>		
Feedback (Negative)	18.022	0.495
Sleep Time	-4.669	0.526
Memory Score	-16.691	< 0.001

Table 2: Results of multiple linear regression models predicting accuracy and response time from feedback type, as well as the covariates sleep time and memory score. The positive feedback group was used as the reference.

In the accuracy model, all predictors failed to reject the null hypothesis as their p-values were all greater than 0.05. For the response time model, however, memory score had a p-value of less than 0.001, which is a lot less than 0.05. Thus the null hypothesis was rejected for memory score, indicating it significantly affected response time. In contrast, feedback type and sleep time both were not significant due to higher p-values, and thus the null hypothesis could not be rejected for those variables.

These results show that within the 2T8 EngSci population, individual cognitive ability—measured by memory score—influences response speed more than feedback type or sleep duration, where a greater memory score leads to less response time as indicated by the negative estimate value.

4.6 Speed-Accuracy Tradeoff



(a) The graph of the Speed-Accuracy tradeoff for the Positive feedback type. The graph is fitted to a linear function (the blue line), with fitted parameters of $m = 0.09 \pm 0.04$ and $b = 14.5 \pm 7.2$. The computed R^2 value is 0.29.

(b) The graph of the Speed-Accuracy tradeoff for the Negative feedback type. The graph is fitted to a linear function (the blue line), with fitted parameters of $m = 0.03 \pm 0.03$ and $b = 18.6 \pm 6.3$. The computed R^2 value is 0.07.

Figure 4: Graphs of Speed-Accuracy tradeoffs across all feedback types.

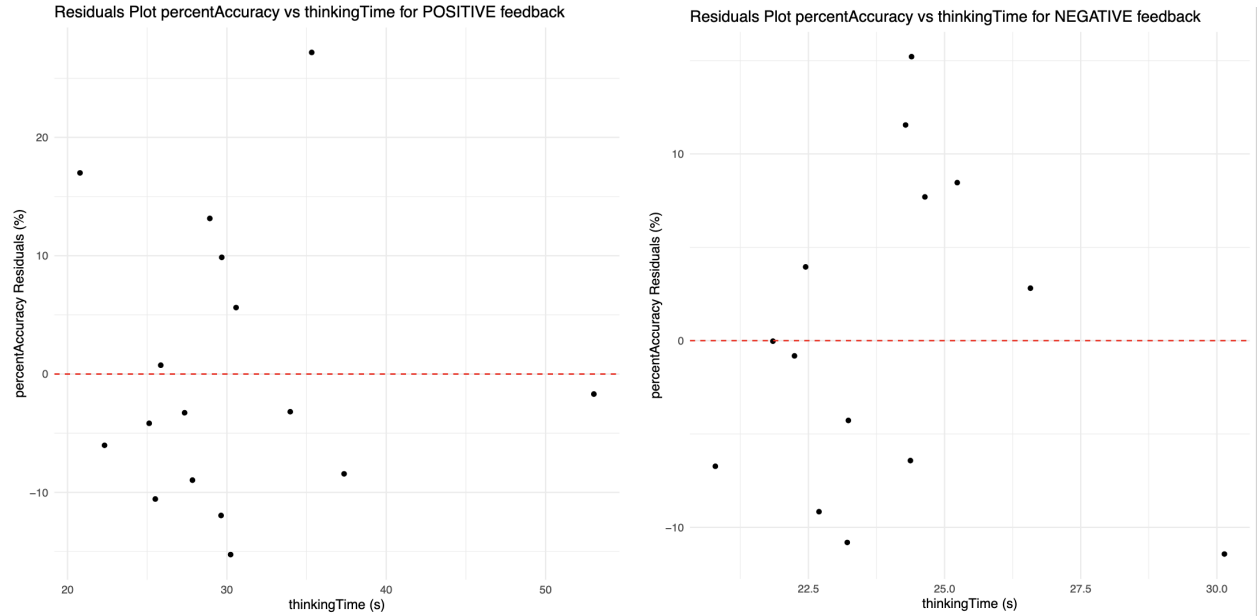
Referencing Figure 4, linear regression was performed on both feedback type cases, with lines of best fit found to be $y = (0.09 \pm 0.04)x \pm (14.5 \pm 7.2)$ on the positive case and

$y = (0.03 \pm 0.03)x + (18.6 \pm 6.3)$ on the negative case, where y is accuracy and x is the thinking time.

From these results, we notice that there seems to be no correlation between thinking time and accuracy for our experiment, as linear regression performs very poorly in fitting to the data. Furthermore, the data visually looks very randomly distributed and noisy, with seemingly no other function that can fit to the data. This is exemplified by a very low R^2 value of 0.29 and 0.07 across Figures 4a and 4b respectively, demonstrating that the linear model only captures a very limited amount of variability in accuracy relative to the mean accuracy.

Additionally, the fit parameter uncertainties (which were calculated from the covariance matrices) are very large relative to the fitted slope parameter, indicating an inaccurate and uncertain fit, further demonstrating poor correlation between thinking time and accuracy.

To further analyze this, a residuals plot of the two time-accuracy plots is shown below:



(a) The residual plot of the Speed-Accuracy tradeoff for the Positive feedback type.

(b) The graph of the Speed-Accuracy tradeoff for the Negative feedback type.

Figure 5: Residual plots of Speed-Accuracy tradeoffs across all feedback types. The black points indicate the residual points $(Y - \hat{Y})$.

Referencing Figures 5a and 5b above, there is also large vertical error in the graphs, where the range of the error extends from about $[-15\%, 27.5\%]$ in the positive case and $[-12.5\%, 15\%]$ in the negative case, demonstrating a bad fit, large error, and therefore low correlation between accuracy and time.

Originally, it was hypothesized that there would be a positive correlation between thinking time and accuracy. Since the game has a discernible pattern, theoretically a person with

more thinking time spent on a problem should increase the accuracy of solving the problem versus randomly attempting inputs until the solution is found. However, upon analyzing the post-study survey, the results showed that 70% of the population figured out the answer via simple memorization and trial and error without realizing the pattern in the answer. Therefore, this is consistent with the results that show a weak relationship and correlation between thinking time and accuracy, as it demonstrates that performance was driven by random strategies over systematic reasoning.

5 Conclusion

The analysis of the means, medians, and quartile ranges of the response time and accuracy proved that positive feedback had a more positive effect on performance than negative feedback. This demonstrates that the initial hypothesis is partly correct, where faster response times were observed, but greater accuracy was also seen. The results of this experiment are likely related to the idea presented in [1], as participants with positive feedback were more likely to be motivated to complete the experiment than participants with negative feedback. The Welch t-test is conducted to verify if there is a statistically significant difference in response time or accuracy due to feedback type. Since the p-values of response time (0.643) and accuracy (0.152) are greater than the $\alpha = 0.05$, the null hypotheses (3, 4) are unable to be refuted, implying the hypothesis cannot be proven as there is insufficient statistical evidence to determine if feedback type affects response time or accuracy in the population of 2T8 Engineering Science students.

Furthermore, upon controlling for covariates, the null hypothesis for feedback type was still unable to be rejected, thus feedback type remains insignificant. Memory score, however, significantly affected response time. This suggests that performance is significantly driven by cognitive ability rather than feedback type within the 2T8 Engineering Science population. The analysis of speed-accuracy tradeoff demonstrate that there is no correlation between response time and accuracy. The data is very noisy, did not seem to fit any general mathematical function (visual inspection, linear regression, and R^2), and had large errors in residual plots. This is likely due to the prevalence of memorization strategies in the experiment and is further shown in the covariates analysis, where memory ability had a great effect.

Some limitations that may explain these results is as follows. Having a small sample size may have limited our ability to detect significant effects of feedback type. Furthermore, the task design may have limited the impact of feedback type, as stated before, most participants used memorization rather than systematic reasoning, reducing sensitivity to feedback differences. A recommendation for future studies is to increase participant count, as well as to design a task that requires more systematic reasoning than memorization.

6 References

- [1] N. Schiavone, E. Cohen, and S. A. McIlraith, “It’s Rational for AI Agents to Procrastinate,” in Workshop on Multi-Agent Systems in the Era of Foundation Models, ICML 2025, 2025.

7 Appendix

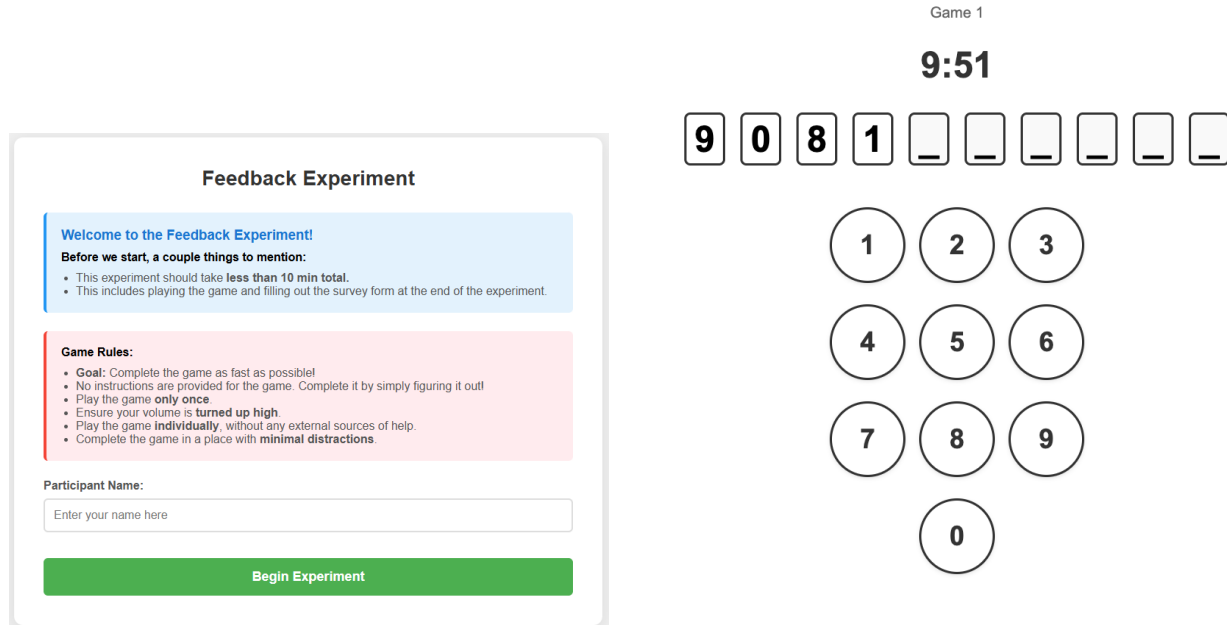
7.1 R Code

The source code for all analysis and graph visualization (ggplot) written in R can be found in this [Github repository link](#).

7.2 Web Application

The web application used to send out to participants and collect data is available [in this link](#).

7.3 Game Interface



(a) Instructions page that the participant sees when clicking on the link to the webpage.

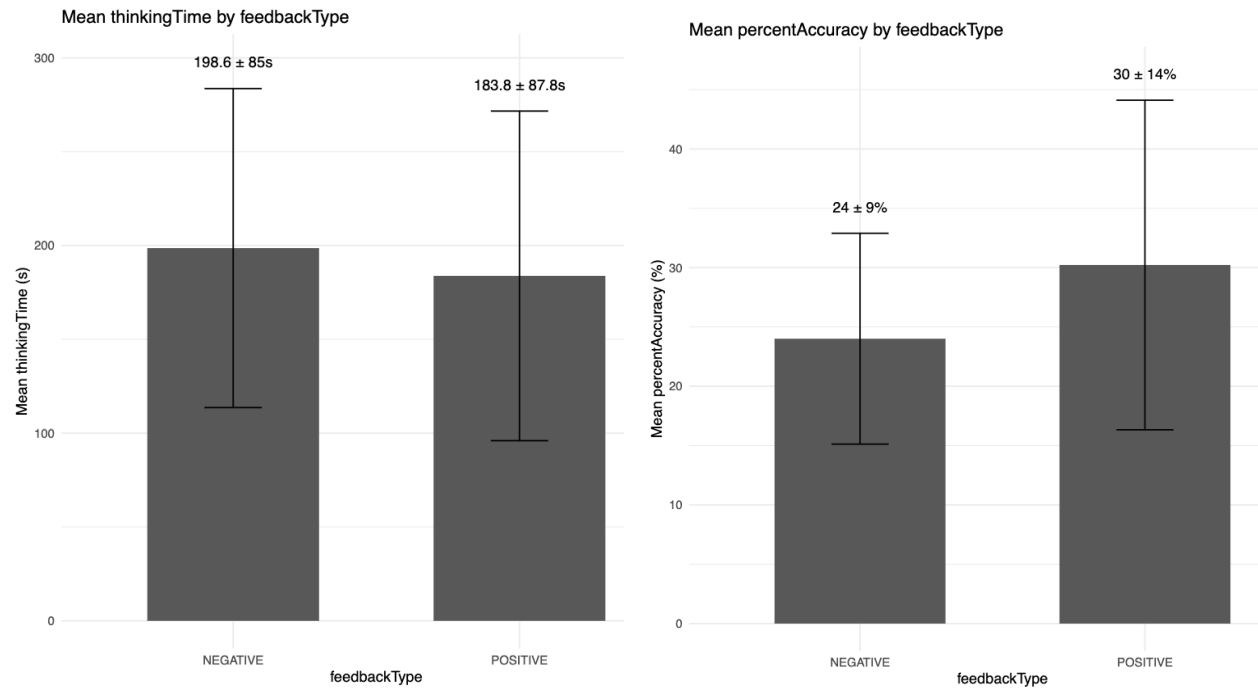
(b) This user interface that the user sees after beginning the experiment.

Figure 6: Experiment apparatus that the user sees when partaking in the experiment.

7.4 CSV Data

The collected data of all 30 participants is available in this [Google Drive link](#).

7.5 Bar Chart of Variable Means



(a) The bar plot of mean thinking time versus feedback types. The mean thinking time for positive and negative feedback is 198.6 ± 85.0 s and 183.8 ± 87.8 s, respectively.

(b) The bar plot of mean accuracy versus feedback types. The mean thinking time for positive and negative feedback is 198.6 ± 85.0 and 183.8 ± 87.8 s, respectively.

Figure 7: Bar plots of thinking time and accuracy across feedback types. Positive feedback yielded a faster thinking time and greater accuracy over negative feedback.